

AHJIN STUDIOS PRODUCTION



Lights, Camera ... Data!

*Analysis By: The
Standard Deviants*





Business Goal

- ☐ To maximize the commercial success of Ahjin Studios' film projects by using industry data to make informed choices about genres, budgets, release timing, and marketing strategy.
- ☐ To estimate the budget costs of creating a film and the ROI.

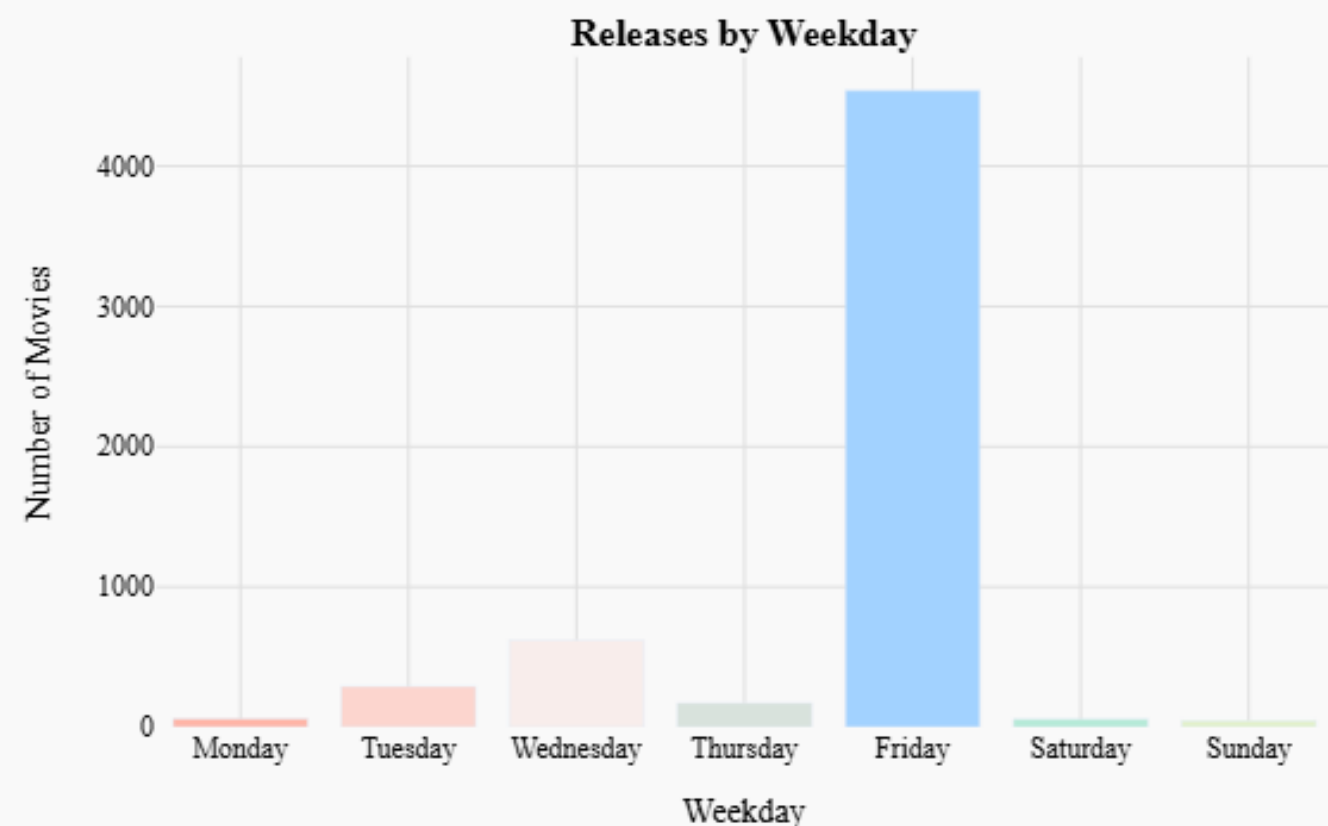
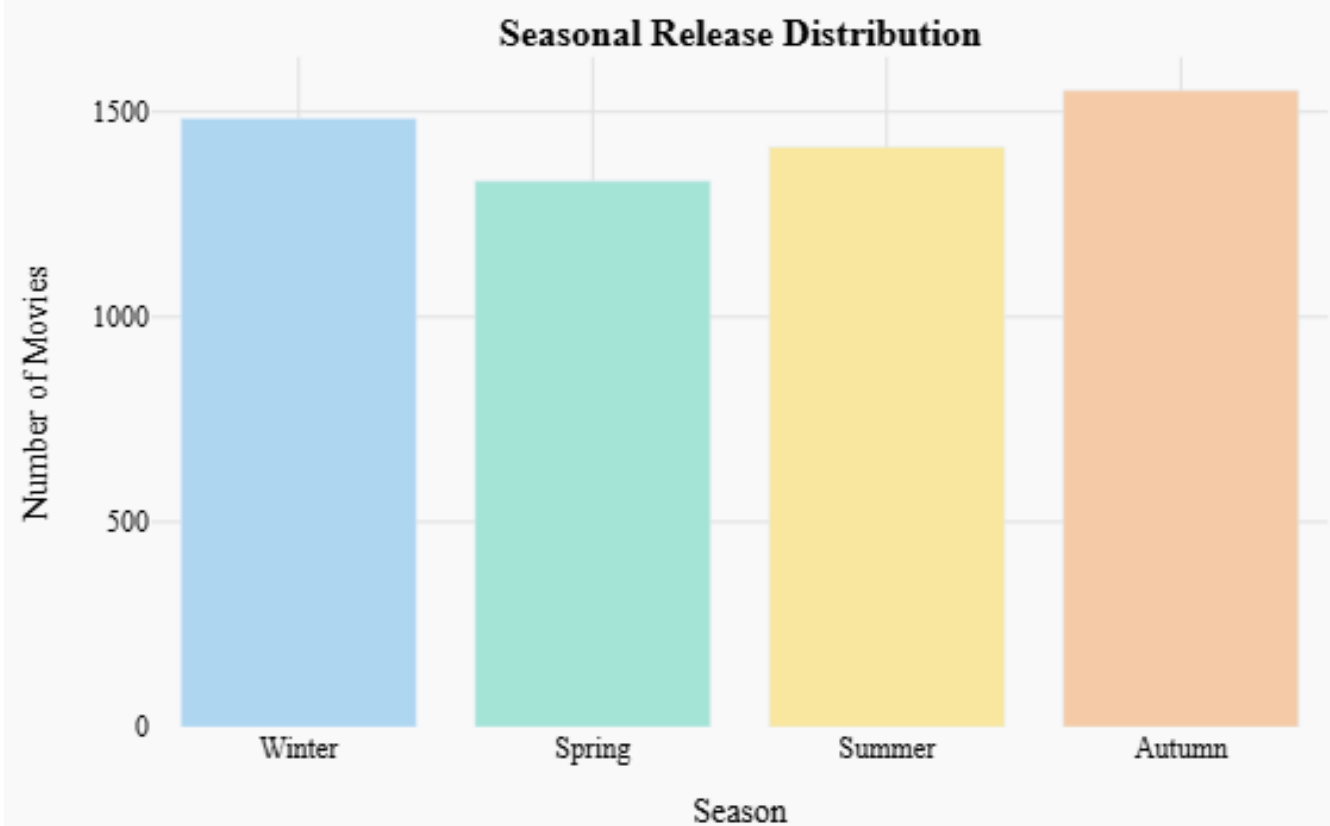
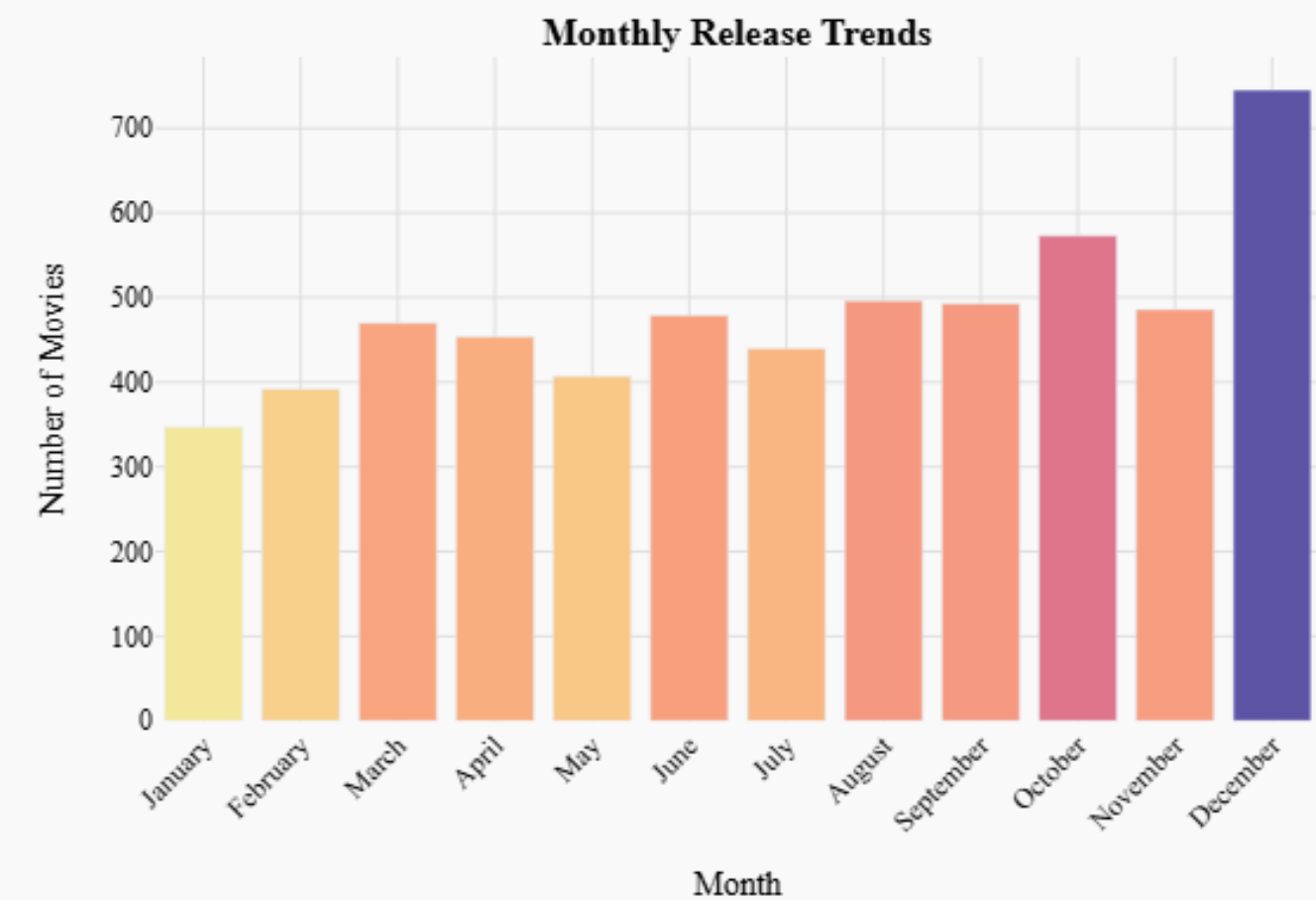
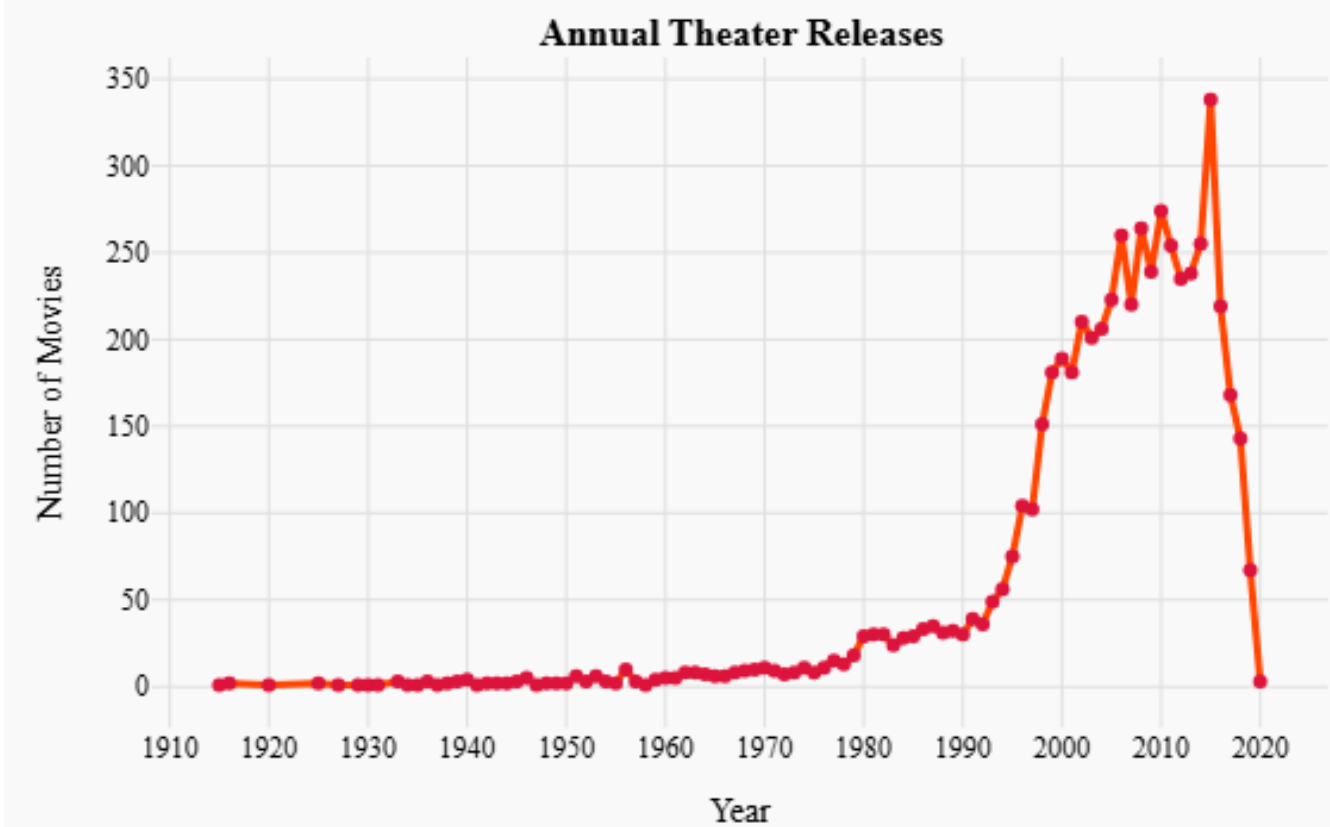
About the Project

- ✧ Ahjin Studios is a fresh face in film, using market insights to analyze recent box office trends and translate key findings into **actionable, data-driven recommendations**, so as to smartly choose and shape movies for success in a tough, ever-changing industry.



WHEN IS THE BEST
TIME TO RELEASE A
MOVIE ?

Movie Release Trends Overview by The Numbers



Observation

Key Trends in Film Releases

Decade Trends

- ❑ Releases surged from the 1980s onward, peaking in the 2000s (>10,000 films).
- ❑ The 2010s saw a slight decline (7,787 films), possibly due to streaming shifts.

Monthly Patterns

- ❑ **Peak Months:** December (3,154), November (3,023), January (2,537) (holiday/awards season).
- ❑ **Summer Blockbusters:** June, July, and August remain strong.
- ❑ **Lowest Month:** February (1,569 films) due to post-awards slump.

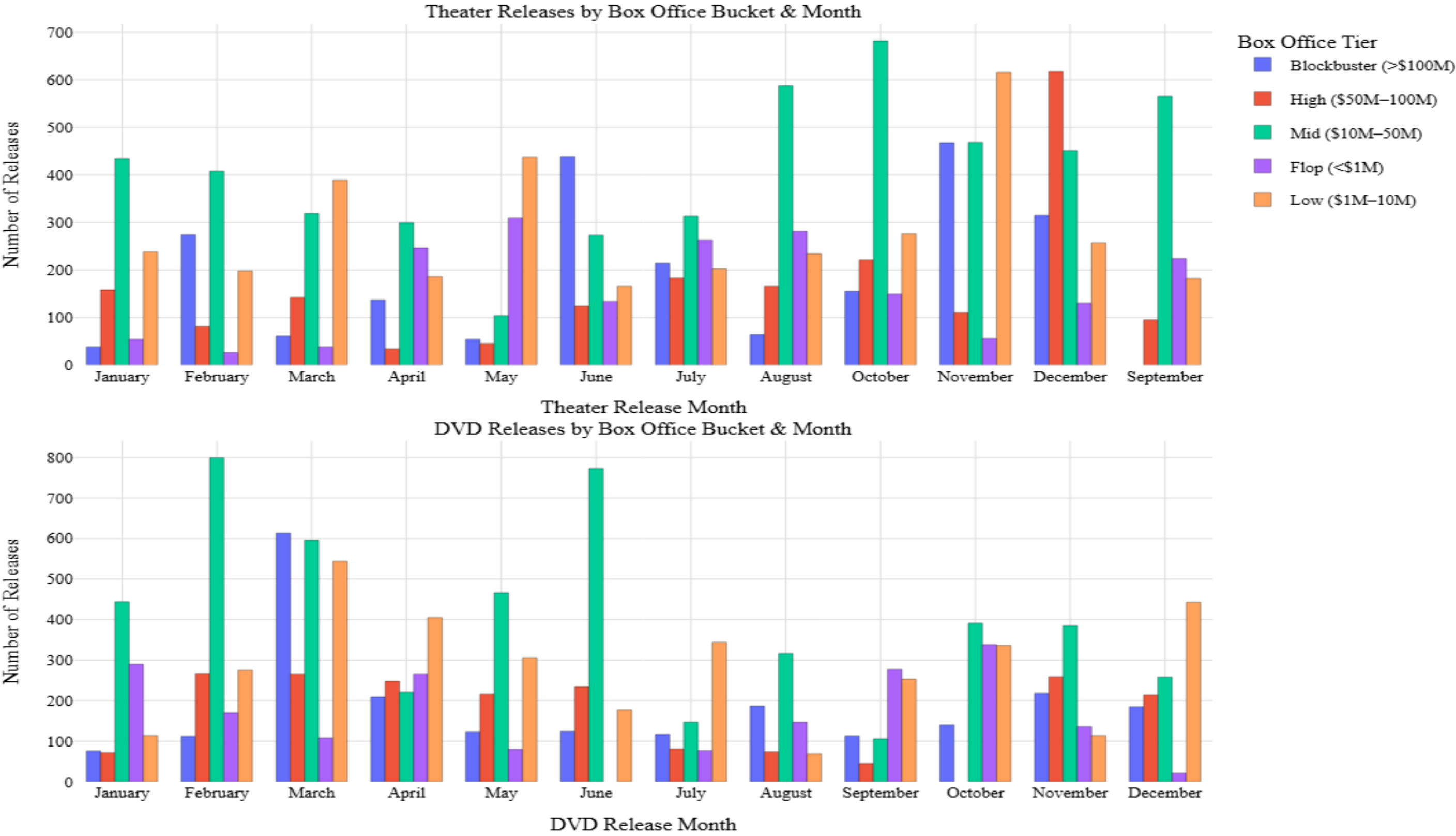
Seasonal Trends

- ❑ **Highest Releases:** Autumn (7,643) and Winter (7,260) dominate (awards/holiday focus).
- ❑ **Lowest Releases:** Spring (5,591) has weaker commercial appeal.

Weekday Releases

- ❑ **Friday** dominates (20,048 films) for weekend box office success.
- ❑ **Midweek (Wednesday: 3,402)** likely for limited/festival releases.
- ❑ **Weekends (Sat/Sun)** are least popular for launches.

Monthly Film Releases: Theater vs DVD by Box Office Tier



Observation

Theater Releases by Box Office Tier

- **Blockbusters (\$100M+)**: Appear consistently but **peak in June, November and December**, aligning with holiday season strategy.
- **Mid-tier (\$10M–\$50M)** and **Low-tier (\$1M–\$10M)** releases hold steady throughout the year, **spiking in October**.
- **High-grossing (\$50M–\$100M)** show strongest in **December**, aligning with awards and holiday family crowds
- **Flops (<\$1M)** surprisingly cluster around **April to July**, perhaps indie or experimental releases.

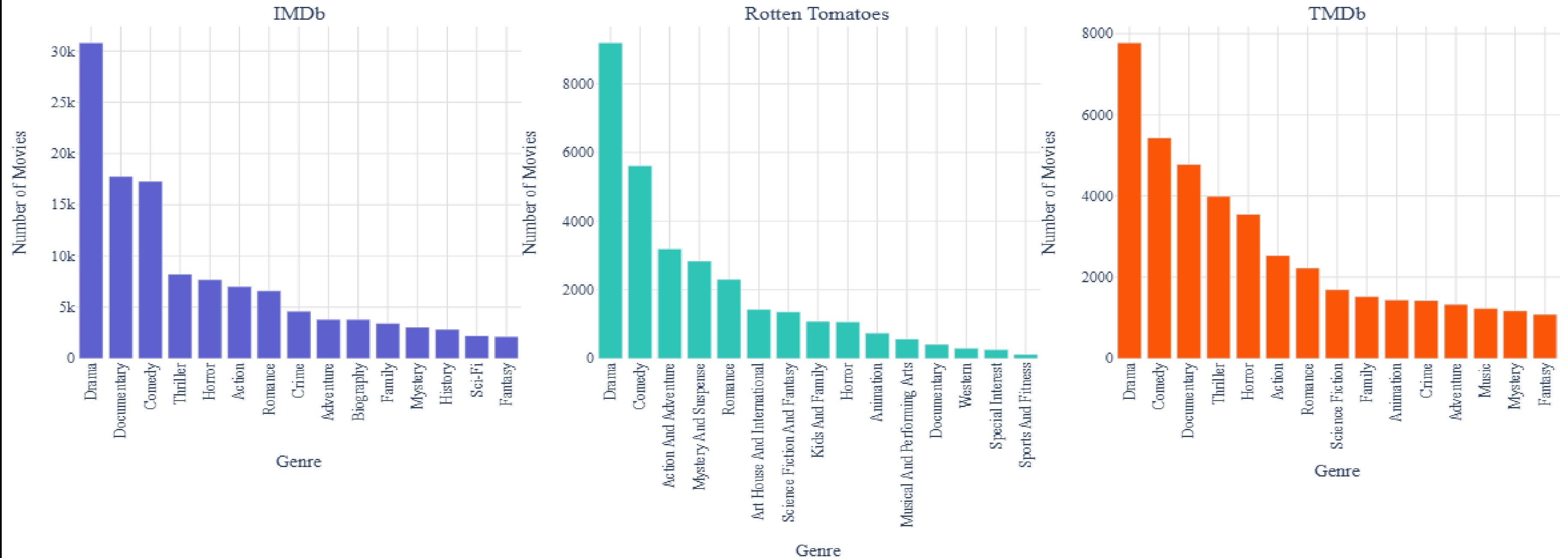
DVD Releases by Box Office Tier

- **Blockbusters and High Grossers** prefer **November–December DVDs**, banking on holiday shoppers.
- **Flops** get DVD drops early in the year - **January to March**- possibly as cleanup from the previous year.
- **Mid & Low grossers** maintain consistent DVD drops, rarely spiking, indicating stable long-tail value.



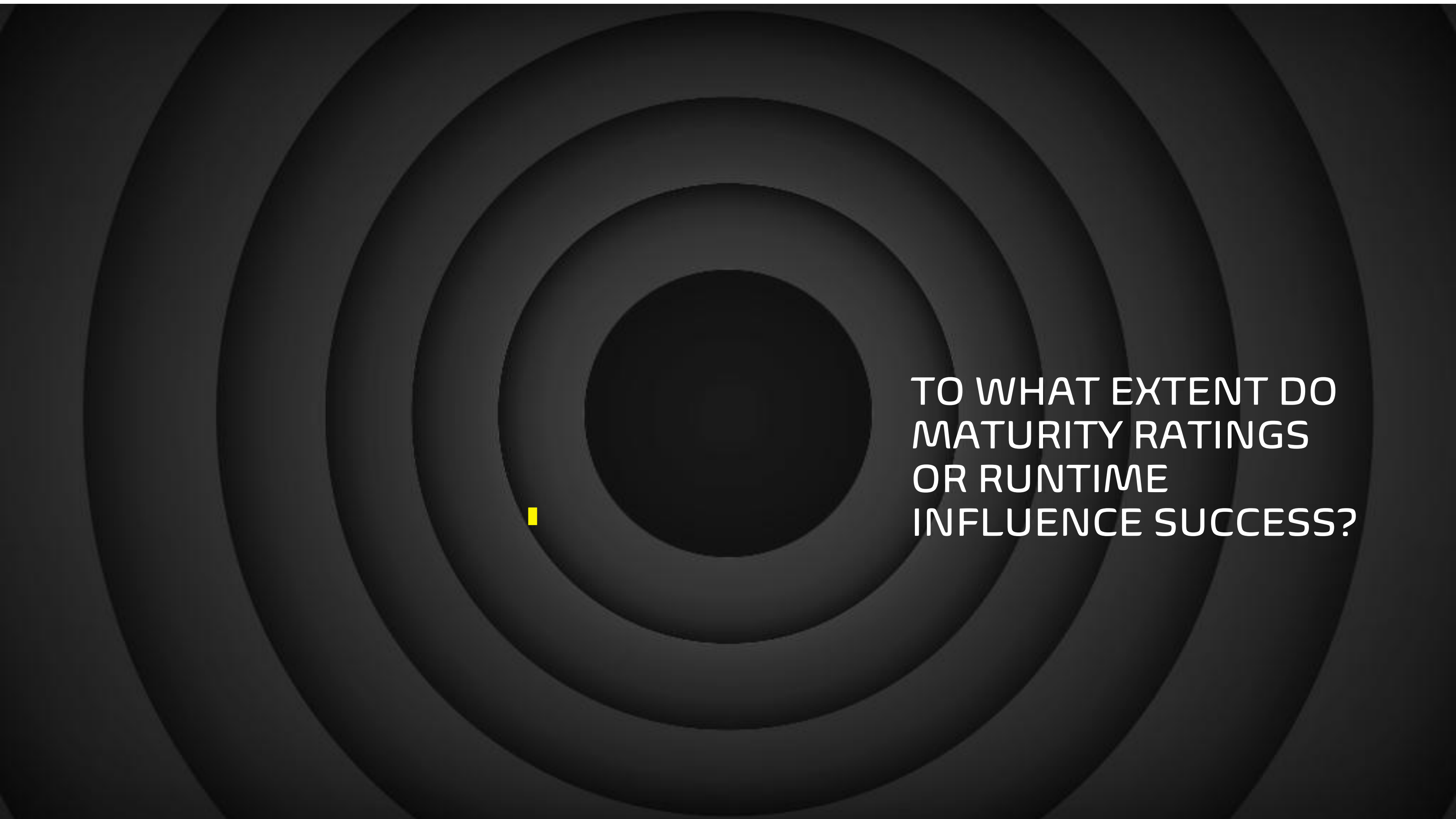
WHAT ARE THE TOP GENRES
ACROSS DIFFERENT
PLATFORMS?

Top 15 Genres by Platform: IMDb vs Rotten Tomatoes vs TMDb



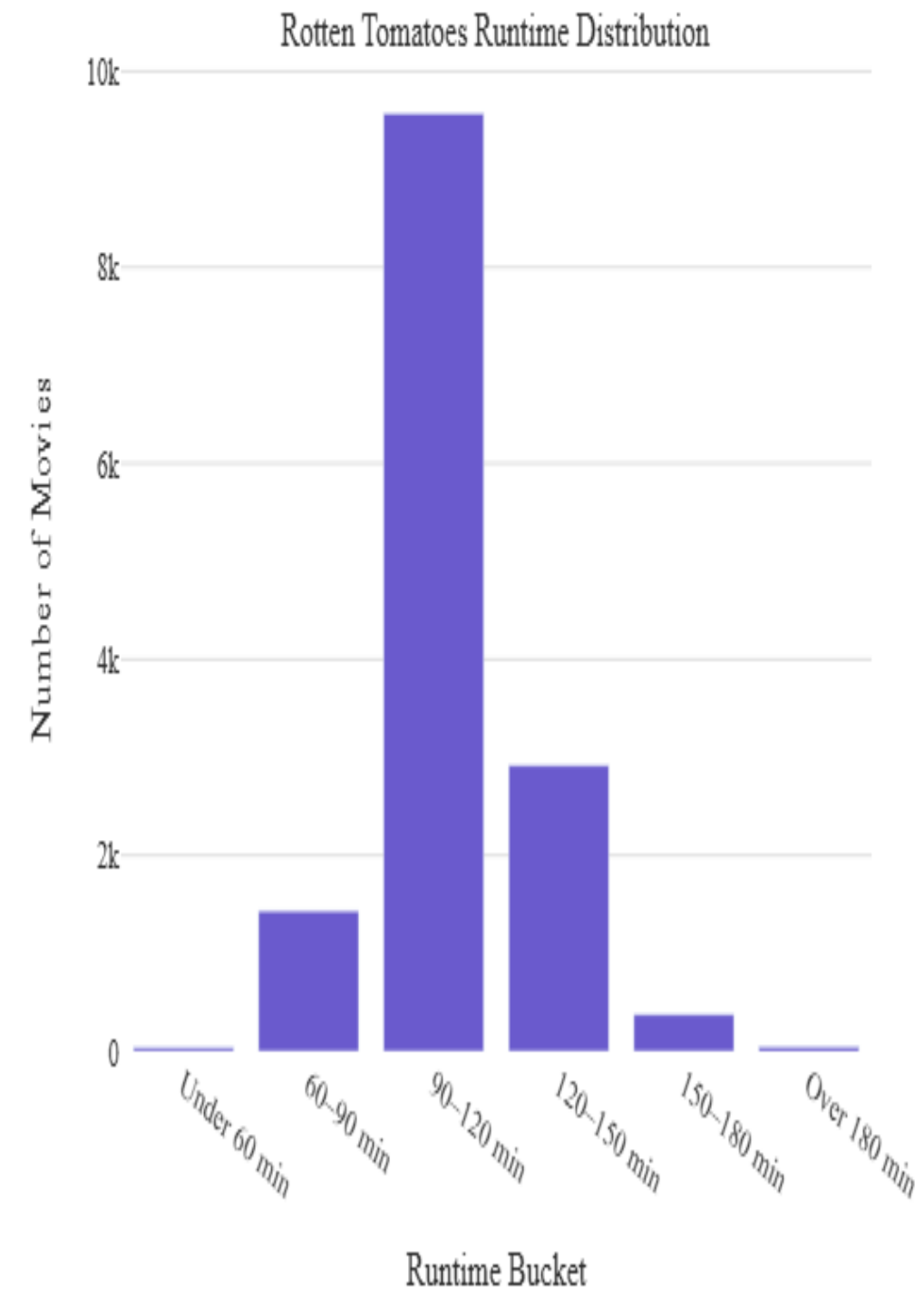
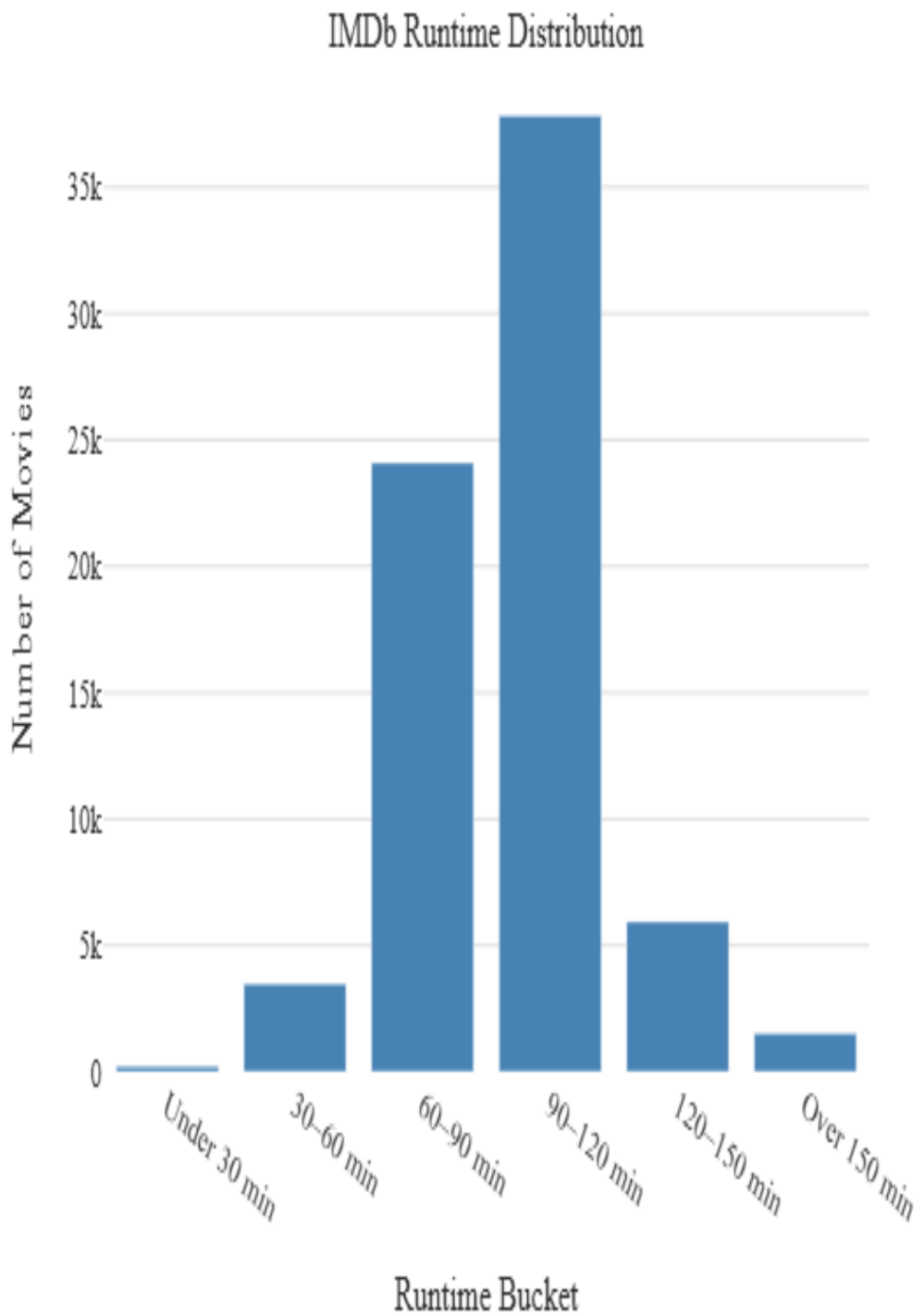
Top 15 Genres: Platform Comparison

- **IMDb:** Clean, standardized genres. *Drama* dominates, followed by *Documentary* and *Comedy*. Structured taxonomy.



TO WHAT EXTENT DO
MATURITY RATINGS
OR RUNTIME
INFLUENCE SUCCESS?

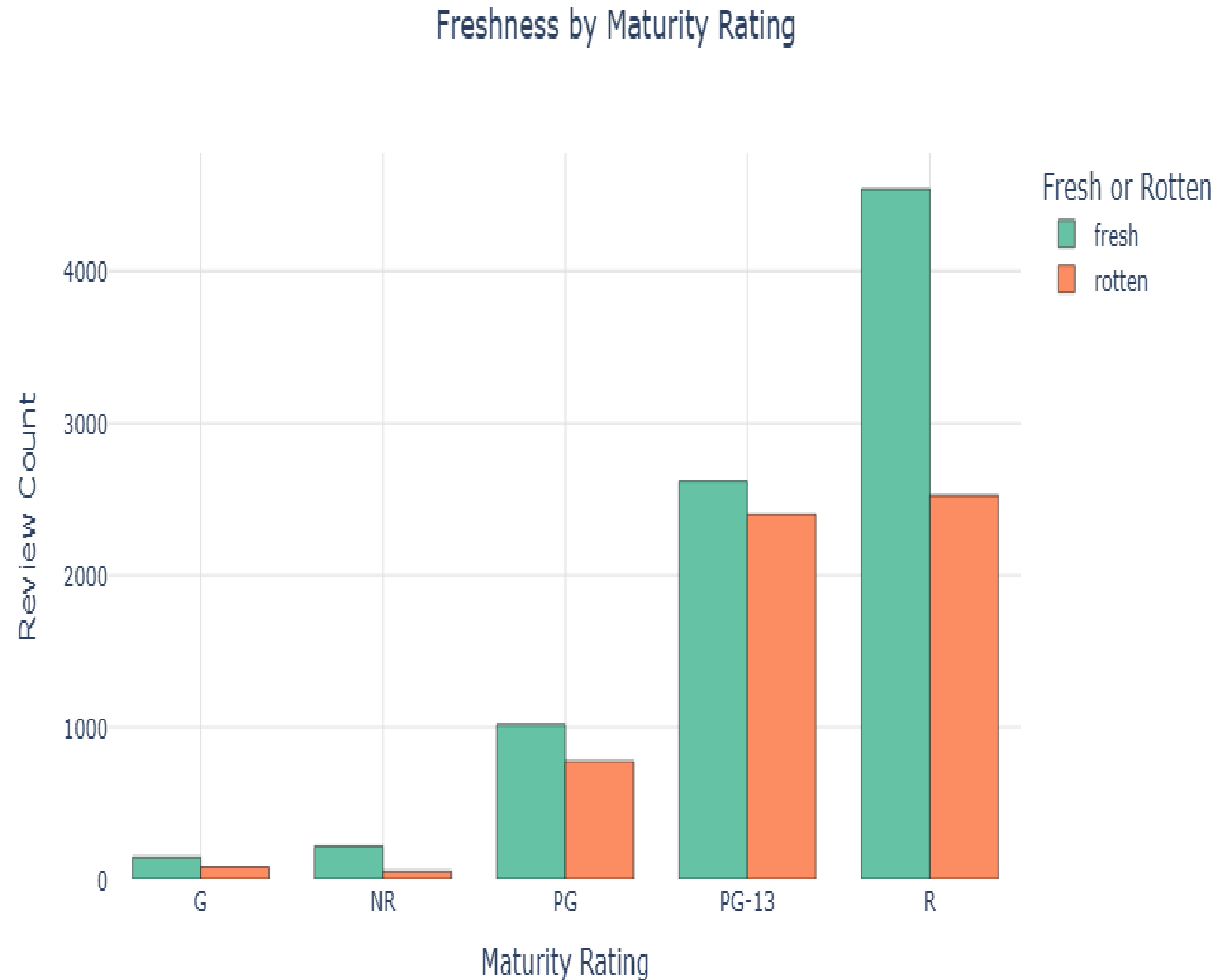
Runtime Distribution Comparison: IMDb vs Rotten Tomatoes



- Across both platforms, **90–120 minutes is the industry sweet spot.**
- IMDb: 37,832 films (largest share)
- Rotten Tomatoes: 18,073 films (also dominant)
- **60–90 minute films are highly represented on IMDb** (24,099 entries), but much less on Rotten Tomatoes—suggesting they’re common, yet not often critically celebrated.
- Shorter formats (under 60 minutes) and ultra-long features (over 150 mins) are **niche segments**—they exist, but rarely drive mainstream buzz or critical acclaim.
- IMDb uniquely records **a significant volume of shorts** (under 30 mins and 30–60 mins), offering insight into experimental and indie trends, but less applicable to theatrical strategy.

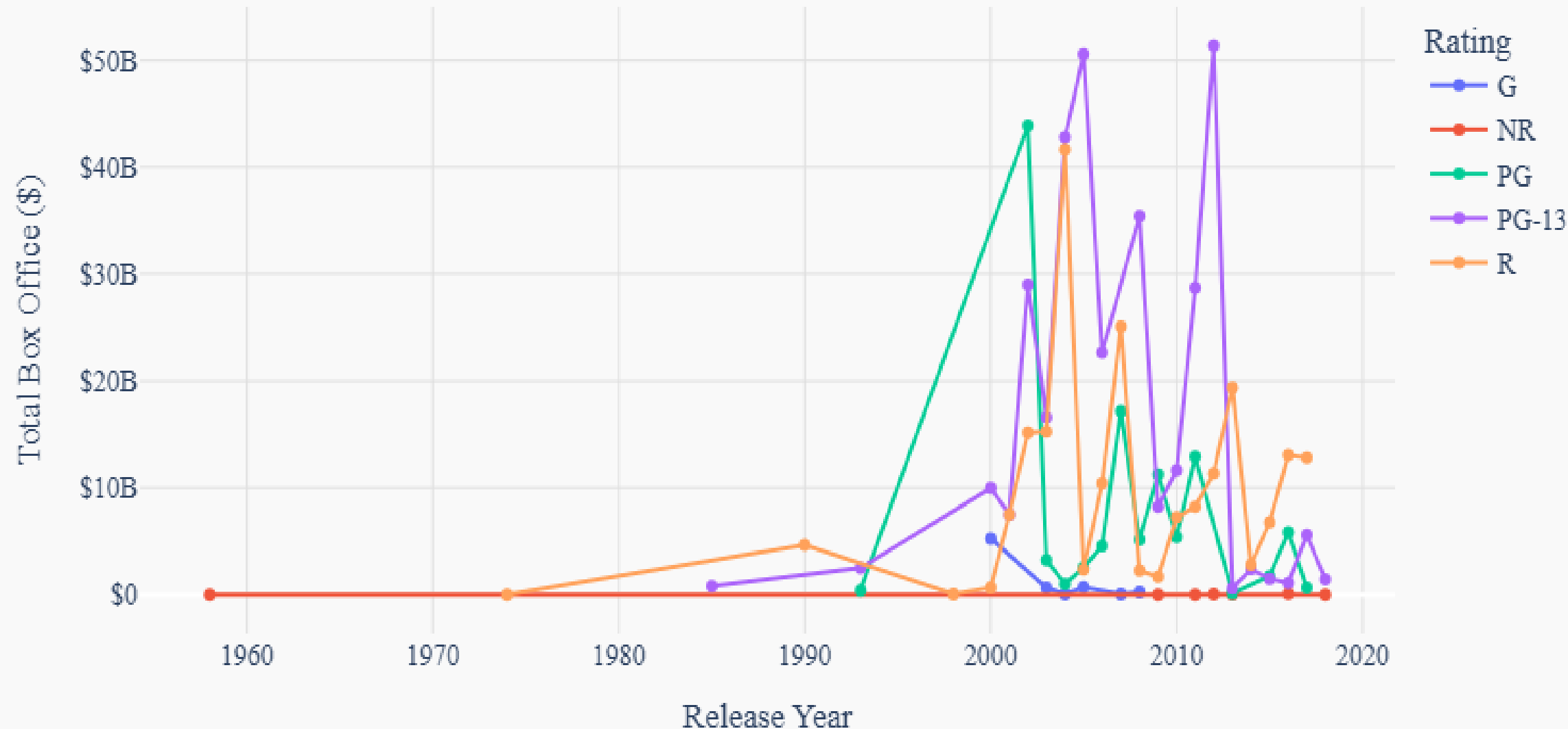
INSIGHT:

- NR is critics' favorite (78.6% Fresh)
- PG-13 floods the market but splits critics.(54.5%)
- R balances mass and quality(63.7% Fresh)
- G/PG = safe but less impactful.(74% & 64% Fresh respectively)



TO WHAT EXTENT DO MATURITY RATINGS OR RUNTIME INFLUENCE SUCCESS?

Box Office Revenue by Maturity Rating (Over Time)



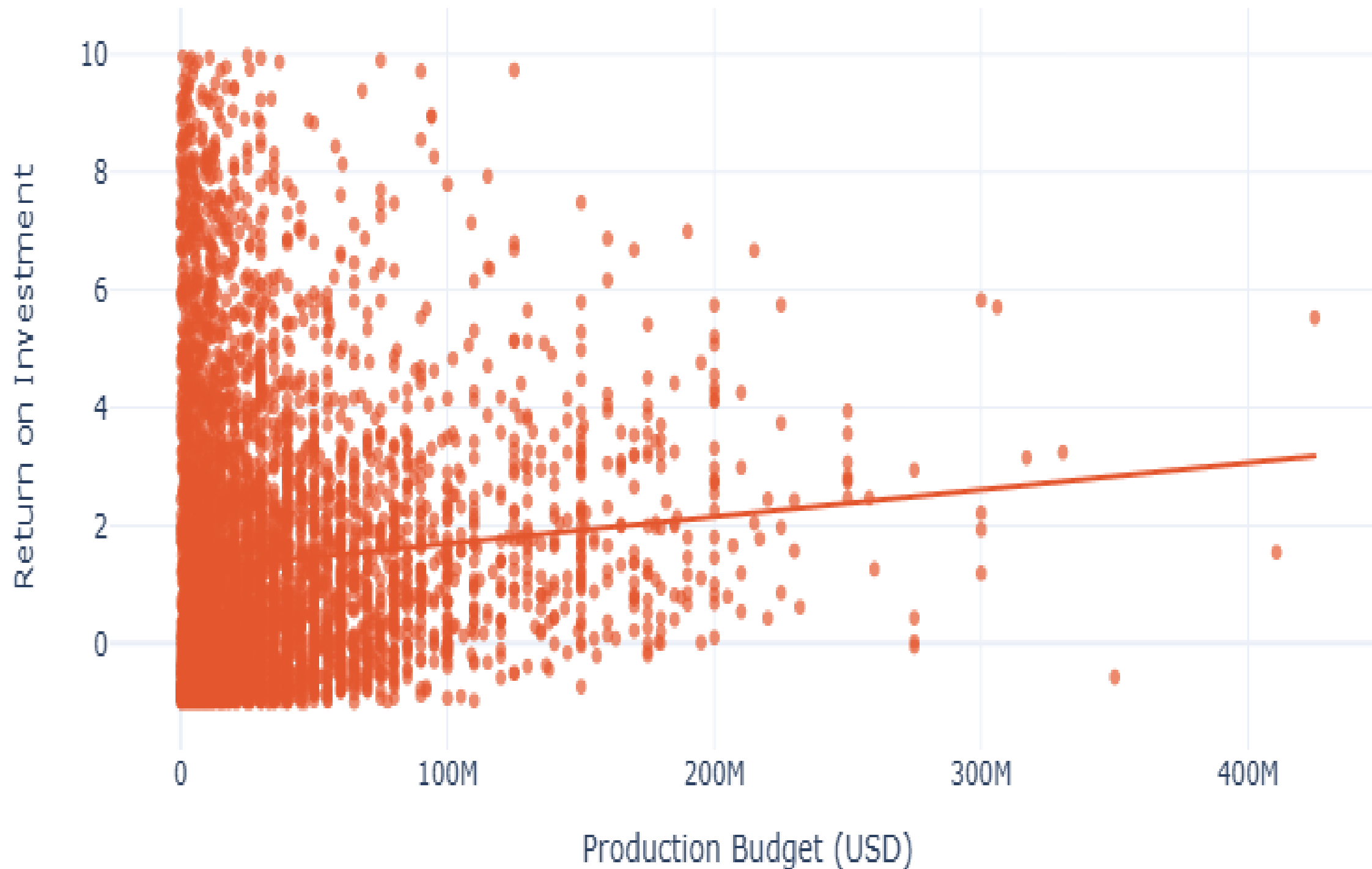
Observation

- **Overall Trend:** The Box Office revenue for all Maturity Ratings has increased significantly since the late 1980s, peaking in the 2000s and early 2010s, before declining in the later years
- **Dominance of PG-13:** The PG-13 rating consistently accounts for the highest box office revenue throughout the period from mid 2000s to the end of the chart, with revenue reaching over \$50 billion around 2010
- **Decline of G and NR:** The revenue for G (general audience) and NR (Not Rated) films remains consistently low throughout the entire period, rarely exceeding a few billion dollars and often staying close to zero



**WHAT BUDGET
RANGE YIELDS
THE BEST ROI?**

Production Budget vs ROI



- **Big Budgets $\neq$ Big Profits**

Spending more gets you bigger box office numbers, **but not better returns.**

Half of all films barely make back their budget.

25% lose over 30% of their investment.

- **Averages Are Misleading**

The "average" movie seems profitable (ROI $\sim 4x$), but that's skewed by rare mega-hits (*Avatar*, *Avengers*).

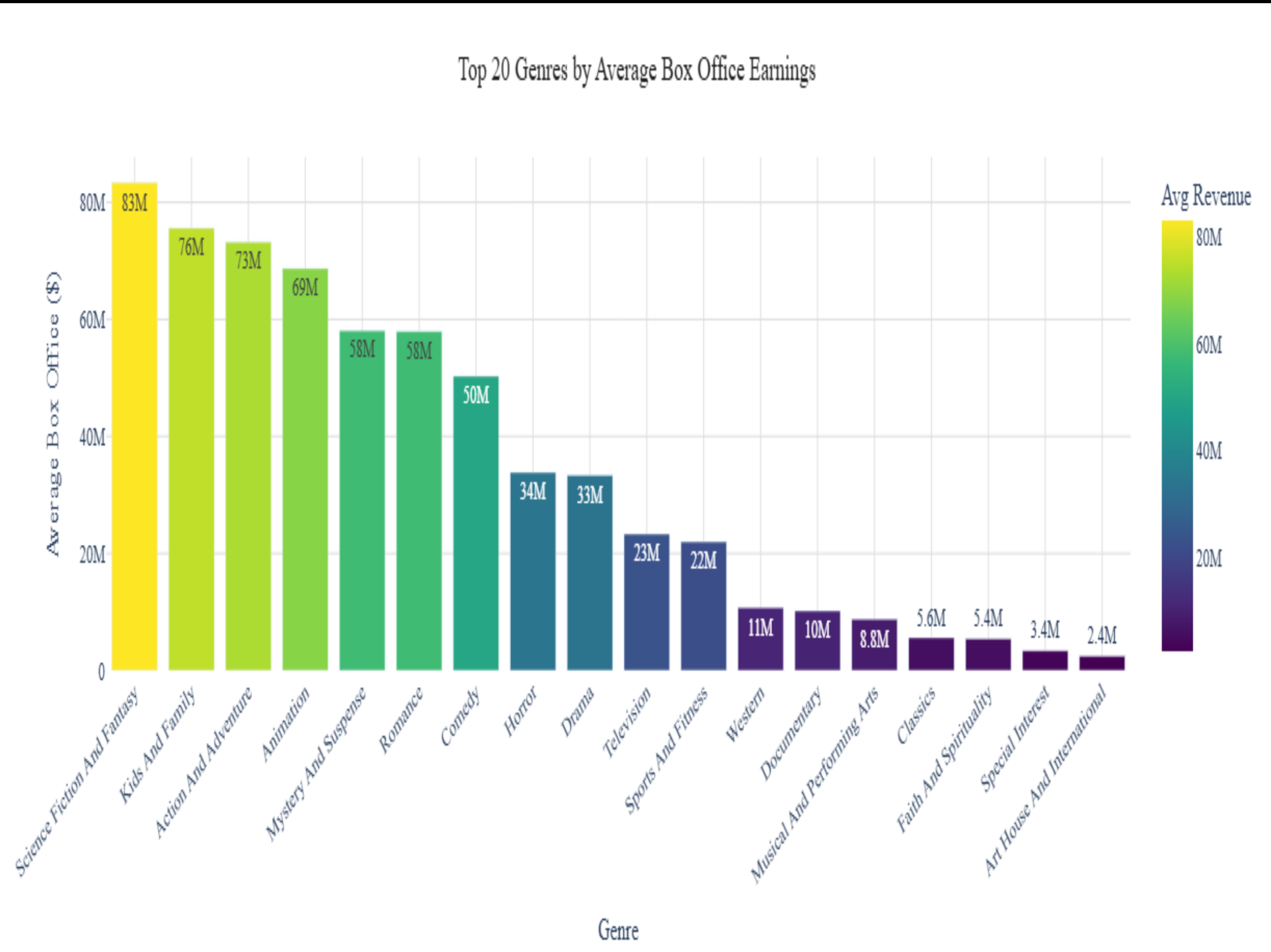
Most films (median ROI = 0.88x) lose money.

- **Small Budgets Punch Harder**

Films under \$20M often deliver **higher ROI** than blockbusters.

Examples: *Paranormal Activity*, *Get Out*, *The Blair Witch Project*.

WHICH GENRES ARE CONSISTENTLY EARNING THE HIGHEST REVENUE?



Observation

Top Dogs:

- Science Fiction & Fantasy leads the pack with the highest average returns—over \$83M.
- Kids & Family, Action & Adventure, and Animation follow closely, all pulling in \$68M—\$75M ranges.

Mid-Tier Successes:

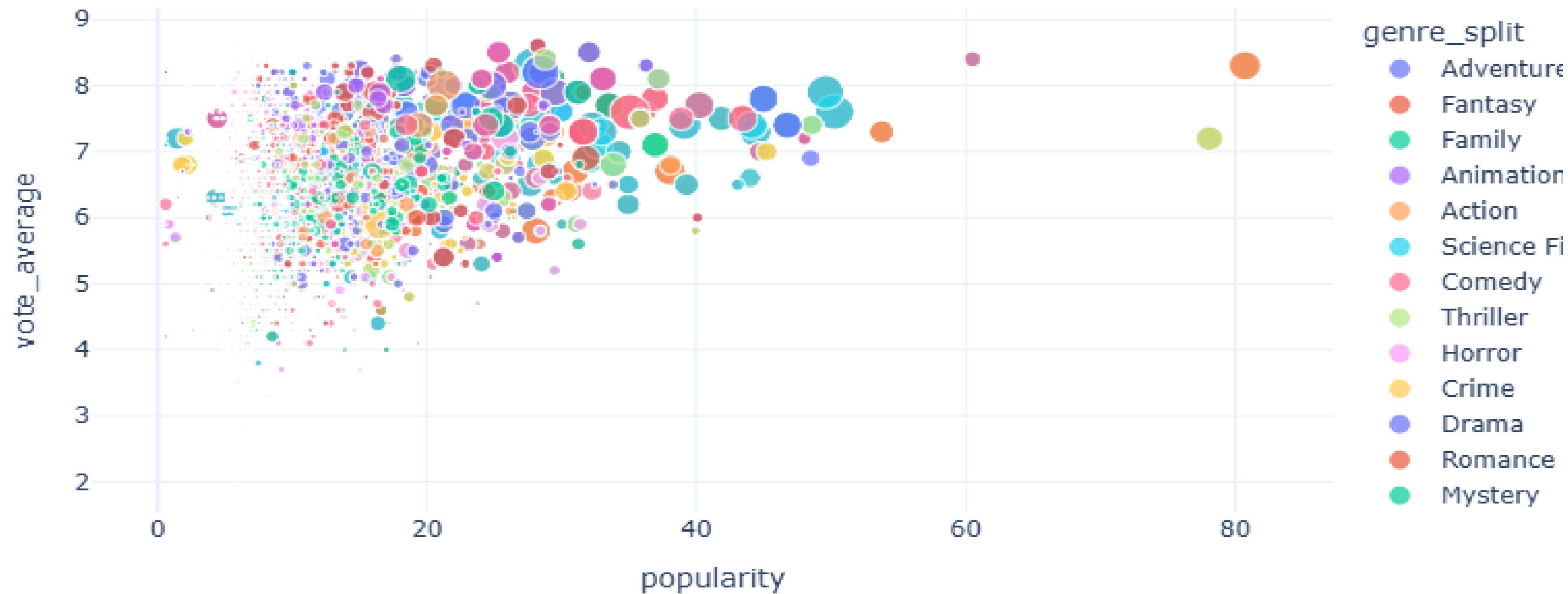
- Mystery & Suspense, Romance, and Comedy hover around the \$50M–\$58M mark. While they’re less spectacle-driven, they benefit from broad relatability, charismatic leads, and cultural timing

Niche and Specialist Fare:

- Genres like Documentaries, Musicals, Faith-based films, Westerns, and Art House see the lowest box office earnings—mostly under \ \$10M.

WHICH GENRES ARE CONSISTENTLY EARNING THE HIGHEST REVENUE?

Genre Popularity against Vote Average by Genre



- At first glance, we see a dense cloud of titles tightly packed between popularity scores of 0–30 and vote averages of 5–8. This is the heartland of cinema—where most films live, breathe, and compete.
- But what’s more telling are the outliers:
 - - A Romance film surges past 80 in popularity, scoring high in votes, showing that the genre can occasionally captivate both the masses and critics.
 - - A Thriller trails closely behind, also claiming a spotlight spot with exceptional popularity and decent ratings.
 - - Science Fiction and Drama genres cluster around the 7.5+ vote average - these genres seem to resonate well with audiences on quality, even if they don’t always dominate popularity.
- Meanwhile, genres like Documentary, TV Movies, and War appear more modest, with generally smaller bubbles and less extreme values - suggesting niche but loyal audiences.

HYPOTHESIS TESTING



Hypothesis 1

Research Question:

Does the maturity rating of a film (G, PG, PG-13, R) significantly affect its box office performance?

Null Hypothesis (H0):

There is **no difference** in the mean box office revenue across different maturity ratings.

Alternative Hypothesis (H1):

At least **one** maturity rating has a **significantly different** mean box office revenue.



Test Chosen : One-Way ANOVA

This test is appropriate because:

The independent variable (maturity rating) is categorical with 4 levels.

The dependent variable (box-office) is continuous.

We are comparing more than two group means simultaneously.

Assumptions:

1. Independent groups: Ratings are independent.

2. Normality: Each group's box office earnings should be roughly normally distributed.

3. Homogeneity of variances: Variance of earnings across groups should be similar.

Decision Rule:

- If $p\text{-value} < 0.05$, reject the null hypothesis. make this brief

Interpretation:

- $F(3, 14107) = 384.31, p < 0.001$, thus the result is highly statistically significant and we can confidently reject the null hypothesis.
- The extremely low p-value (< 0.0001) confirms that differences in box office revenue across maturity ratings are not due to chance.
- Therefore, **Maturity Rating has a statistically effect on box office revenue**

Strategic Insight for Ahjin Studios:

- ❖ **PG-13 and PG films** outperform **G and R-rated** films by **over 2x** in average revenue.
- ❖ For a new studio without franchise leverage or global name recognition, targeting a **PG or PG-13 rating** offers the **broadest demographic reach** (including teens and families) and **highest ROI Potential**.
- ❖ Avoid debuting with **R-rated content** unless aiming for niche prestige or festival circuits; the market signals lower revenue ceiling.
- ❖ **G-rated** films, while safe, may be **overly limiting** unless operating in the animation/kids content niche.

Recommendation:

- ❖ Ahjin Studios should **prioritize greenlighting PG or PG-13 rated films** for theatrical releases to maximize commercial success, especially in its initial production phase.
- ❖ These maturity brackets not only command the highest mean revenues but also align best with the global box office's appetite for broad-access storytelling.



HYPOTHESIS 2

Business Rationale:

- ❑ Ahjin Studios does not have the financial leverage to compete with \$100M+ blockbuster productions. To de-risk its market entry, it must identify the budget tier that offers the best trade-off between investment and return.

This hypothesis evaluates whether:

- ❑ **Mid-budget films**(\$10M–\$30M) deliver **significantly higher ROI** than their high-budget counterparts (>\$70M), making them a strategic sweet spot.

Null Hypothesis (H0):

- ❑ There is **no significant difference** in ROI between mid-budget and high-budget films.

Alternative Hypothesis (H1):

- ❑ **Mid-budget films** have a **significantly higher** mean ROI than high-budget films.

Test Chosen: Independent Samples T-Test (One-Tailed)

Groups:

- -Mid-Budget(production_budget between \$10M & \$30M)
- -High-Budget(production_budget above \$70M)

Dependent Variable: ROI

Test Type: One-tailed t-test to detect whether ROI is statistically greater in the mid-budget group.

Decision Rule:

- If **p-value < 0.05**,reject H0 - conclude that mid-budget films perform better.
- If **p-value $\geq$ 0.05**, fail to reject H0 – no evidence mid-budget films are more profitable.

Interpretation

A linear regression model was fit to assess how budget tier influences return on investment (ROI).

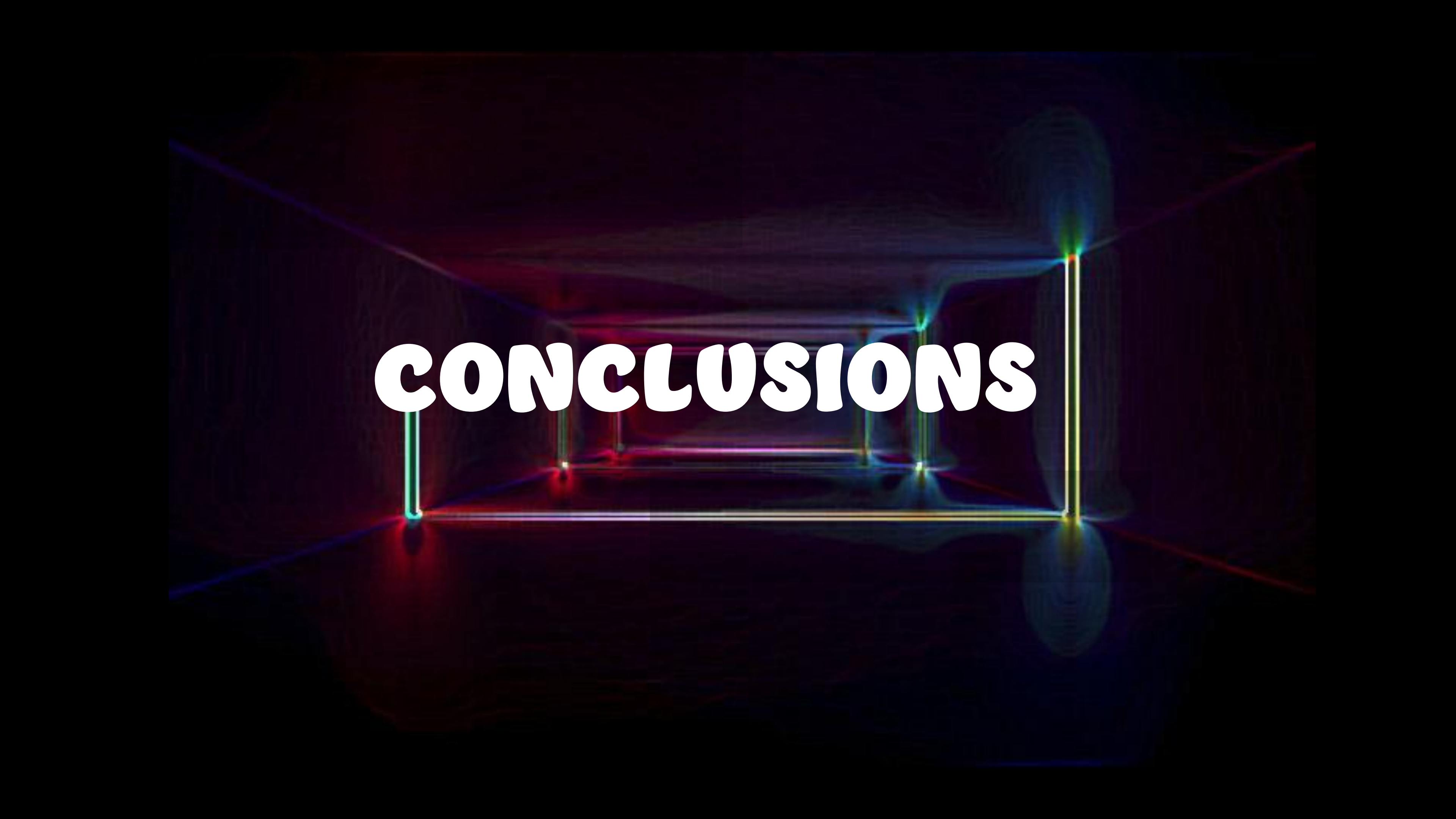
$R^2 = 0.008$ - While the model explains a small proportion of ROI variance, the difference in returns between tiers is still statistically meaningful.

What this means:

- Mid-budget films do not statistically outperform high-budget films in ROI. With a p-value of 0.649, there is no significant evidence to support the hypothesis.
- Low-budget films deliver significantly better ROI, reinforcing the high-risk/high-reward potential of lean productions.
- This finding suggests that profitability in film is less about spend, and more about efficiency, audience targeting, and potentially distribution strategy.

Recommendation

- Prioritize low-budget films (< \$10M) to maximize capital efficiency and minimize downside risk. Strong creative execution can still yield exceptional ROI.
- Re-evaluate the assumption that mid-budget films are a safer bet - the data doesn't support this claim. Without strong differentiators (e.g. IP, genre innovation), mid-budget investments may underperform.

The background is a dark, deep blue space filled with vibrant, multi-colored light trails in shades of red, green, and yellow. These trails form a series of concentric, perspective-driven rectangular frames that recede into the distance, creating a strong sense of depth and movement. The word "CONCLUSIONS" is centered in the middle of these frames.

CONCLUSIONS

Strategic Summary for Ahjin Studios

1. Release Timing Matters

- - Nov-Dec : Prime window for major releases (holiday audiences, awards consideration).
- -Deploy mid-budget or high-concept films where competition is thinner but attention is available - especially for thrillers, dramas, or genre hybrids.

2. Runtime = Perceived Value

- - 150-180 min films (PG/PG-13) perform best—seen as "event cinema."
- - Avoid <90 min unless stylistically intentional (risk of seeming low-value).

3. Target PG/PG-13 for Commercial Viability

- - Broadest audience appeal (family-friendly, international markets).
- - Reserve R-rated for prestige/awards plays only.

4. Genre Strategy: Blend for Impact

- - High-ROI genres :Thrillers, mysteries, dramas (lower cost, strong returns).
- - Avoid VFX-heavy blockbusters until scale justifies risk.
- - Hybrid narratives (e.g., thriller + social commentary) maximize engagement.

5. Budget with Precision, Not Excess

- - Big budgets ≠ guaranteed profits(many flops are overfunded).
- - Tiered funding: Match budgets to genre, talent, and market potential.
- - Data-driven greenlights: Assess ROI potential early (script stage).



RECOMMENDATIONS

RECOMMENDATIONS

In Summary:

1. Build around PG/PG-13 narratives with emotional scope
2. Lean into genre smart , budget-disciplined storytelling
3. Let runtime, tone and genre be strategic signals, not creative afterthoughts





THANK YOU!